

Literature Status: Daws–Runde’s $B(\ell^p)$ Amenability Question

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2026-07-18

Status

Status: literature already answered. The source paper is Matthew Daws and Volker Runde, *Can $B(\ell^p)$ ever be amenable?*, arXiv:0711.4311, *Studia Mathematica* 188 (2008), 151–174. The supporting answer is Volker Runde, *$B(\ell^p)$ is never amenable*, arXiv:0907.3984, *J. Amer. Math. Soc.* 23 (2010), 1175–1185.

Source Question

The abstract and introduction of arXiv:0711.4311 record the then-open problem:

is $\mathcal{B}(\ell^p)$ amenable for $p \in (1, \infty) \setminus \{2\}$?

The paper proves consequences of a hypothetical positive answer. In particular, it shows that amenability of $\mathcal{B}(\ell^p)$ would imply amenability of $\ell^\infty(\mathcal{B}(\ell^p))$ and $\ell^\infty(\mathcal{K}(\ell^p))$, and it develops a dichotomy for $\ell^\infty(\mathcal{K}(E))$ over infinite-dimensional $\mathcal{L}^p$ -spaces.

Answering Paper

Runde’s later paper arXiv:0907.3984 explicitly addresses this same line of questions. Its abstract states that

$$\ell^\infty(\mathcal{K}(\ell^2 \oplus E))$$

is not amenable for a class of spaces including $E = \ell^p$, $p \in (1, \infty)$. It further states the consequence that $\ell^\infty(\mathcal{K}(E))$ is not amenable for every infinite-dimensional $\mathcal{L}^p$ -space E , and hence that $\mathcal{B}(\ell^p(E))$ is not amenable for any $\mathcal{L}^p$ -space E . Taking $E = \mathbb{C}$ gives non-amenability of $\mathcal{B}(\ell^p)$ for the open range from the source paper.

Scope

This clears the main open problem in arXiv:0711.4311 and the main $\ell^\infty(\mathcal{K}(E))$ side frontier for $\mathcal{L}^p$ -spaces. It does not settle Johnson’s general problem asking whether $\mathcal{B}(E)$ can be amenable for some infinite-dimensional Banach space E , nor the fully general abstract question whether amenability of $\ell^\infty(A)$ forces amenability of A^{**} with an Arens product.

Run Note

The run already contains a related literature-status packet for Ozawa's earlier formulation. This packet records the direct answer relation for the later Daws–Runde source and its stronger $\ell^\infty(\mathcal{K}(E))$ side questions.